

Simple Stresses and Strains

Strength of Materials - Unit 2: Simple Stresses and Strains - Session 1

1. Stress

When an external force acts on a body, internal resisting forces develop. **Stress (σ)** is the internal resisting force per unit area: $\sigma = F / A$. Its SI unit is the pascal ($1 \text{ Pa} = 1 \text{ N/m}^2$); engineers usually work in MPa, where $1 \text{ MPa} = 1 \text{ N/mm}^2$.

- **Tensile stress:** the force pulls the body and tends to lengthen it.
- **Compressive stress:** the force pushes the body and tends to shorten it.
- **Shear stress (τ):** the force acts parallel to the area, $\tau = F / A$.

2. Strain

Strain (ϵ) is the deformation per unit original length: $\epsilon = \Delta L / L_0$. Being a ratio of two lengths, strain has no unit; it is often quoted in microstrain (10^{-6}).

3. Hooke's law and Young's modulus

Within the elastic limit, stress is directly proportional to strain:

$$\sigma = E \epsilon$$

The constant E is the modulus of elasticity (Young's modulus). A larger E means a stiffer material: it stretches less under the same stress.

Material	Young's modulus E (GPa)	Typical use
Structural steel	200	beams, columns, rods
Aluminium alloy	70	aircraft frames
Copper	117	electrical conductors
Concrete	30	foundations, slabs
Timber (along grain)	12	roof trusses

4. Stress-strain curve of mild steel

A tensile test pulls a standard specimen until it breaks. The curve shows: the **proportional limit** (end of the straight line), the **elastic limit** (largest stress with full recovery), the **yield point** (strain grows with little extra stress), the **ultimate tensile strength** (maximum stress) and finally **fracture** after necking.

5. Worked example

A steel rod 2 m long and 20 mm in diameter carries an axial tensile load of 50 kN. Take $E = 200 \text{ GPa}$. Find the stress, the strain and the elongation.

1. Area: $A = \pi d^2 / 4 = \pi \times 20^2 / 4 = 314.16 \text{ mm}^2$.
2. Stress: $\sigma = F / A = 50\,000 \text{ N} / 314.16 \text{ mm}^2 = 159.15 \text{ N/mm}^2 = 159.15 \text{ MPa}$.
3. Strain: $\epsilon = \sigma / E = 159.15 / 200\,000 = 7.96 \times 10^{-4}$.
4. Elongation: $\Delta L = \epsilon L_0 = 7.96 \times 10^{-4} \times 2000 \text{ mm} = 1.59 \text{ mm}$.

The stress is below the yield stress of mild steel (about 250 MPa), so the rod returns to its original length when the load is removed.

6. Common misconceptions

'Stress is the same as force': the same force produces a much larger stress on a thin rod than on a thick one. 'A stiffer material is always stronger': stiffness (E) and strength (ultimate stress) are different properties; glass is stiff but brittle, while rubber is flexible but tough.

7. Practice

1. An aluminium bar 1.5 m long with a 25 mm $\times$ 25 mm square section carries a tensile load of 40 kN. Take $E = 70 \text{ GPa}$. Find the stress, strain and elongation.
2. Why is strain dimensionless while stress is not?